

# Code Summary & Advantages

## chat\_with\_pdf.py

🔥 **Summary:** This script builds a Streamlit web app for user interaction, likely for health or chatbot purposes.

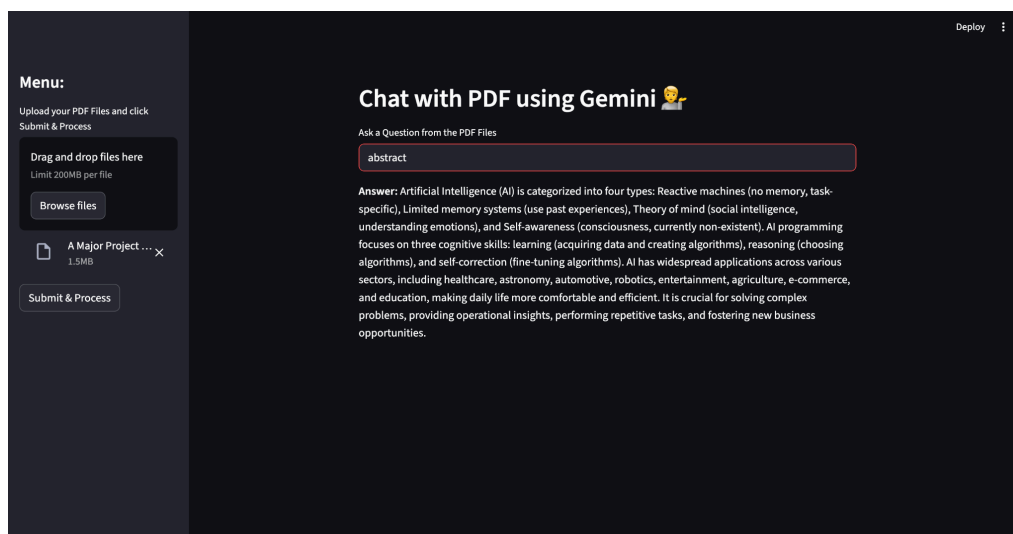
✅ **Advantages:**

- Reusable functions and modular code structure.
- Integration with modern libraries (Streamlit, Gemini LLM, PyPDF).
- Facilitates interactive AI-driven workflows.

## chat\_with\_pdf.py (Chat with PDFs using Gemini + LangChain)

- This is a **Streamlit app** where you upload PDFs and then **ask questions about them**.
- It works like a **chatbot for your PDF documents**.
- Steps it follows:
  1. **Upload PDFs** → Extracts all text from them.
  2. **Splits text into chunks** → so AI can handle large documents.
  3. **Stores chunks in FAISS database** → (a search index).
  4. When you ask a **question**, it:
    - Searches relevant chunks from the PDF.
    - Sends those chunks + your question to Gemini AI.
    - Returns an **answer only based on the PDF content**.
- If the answer isn't in the PDF, it replies with **"I don't know."**

👉 In short: **Upload PDF** → **Ask question** → **AI finds answer inside PDF** → **Shows you**.



## rag\_with\_gemini\_LLM.py

🔗 **Summary:** This script builds a Streamlit web app for user interaction, likely for health or chatbot purposes.

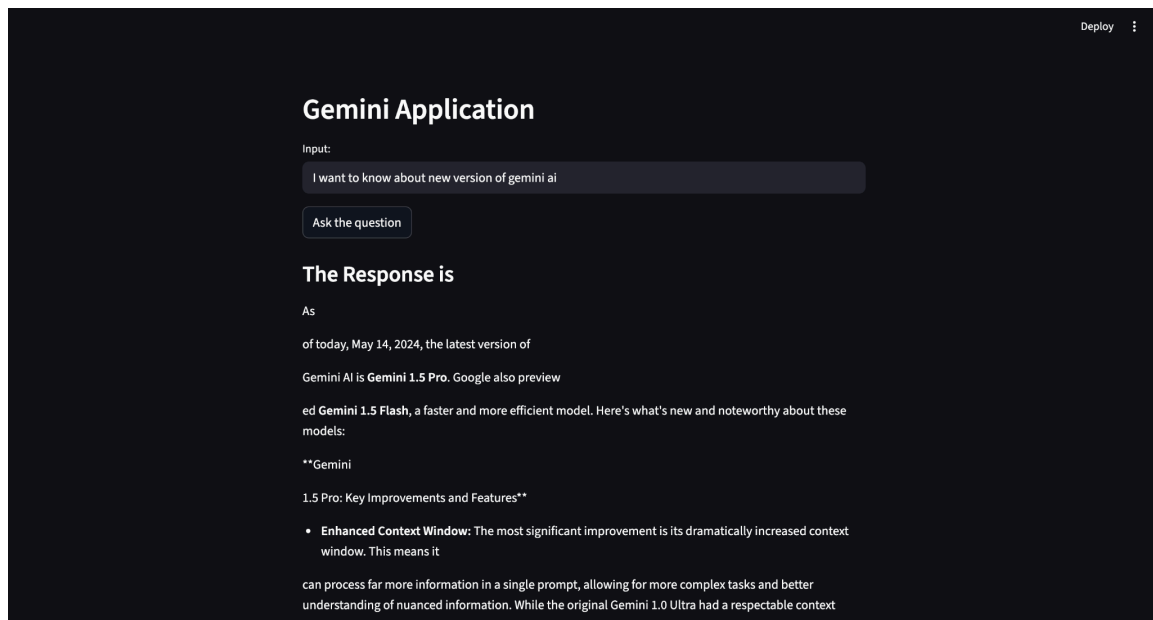
### ✅ **Advantages:**

- Reusable functions and modular code structure.
- Integration with modern libraries (Streamlit, Gemini LLM, PyPDF).
- Facilitates interactive AI-driven workflows.

## rag\_with\_gemini\_LLM.py (Q&A Chatbot with Gemini AI)

- This is a **Streamlit chatbot app** using **Gemini AI**.
- You type any question, and Gemini replies in real time.
- Key points:
  - Uses **Gemini 2.0 Flash model**.
  - Keeps a **chat history** (remembers past Q&A).
  - Streams the answer piece by piece (so you see it as it generates).
  - Simple interface → just a text box + button.

👉 In short: **Ask any question → Gemini replies instantly → Keeps chat memory.**



## calories\_health.py

🔴 **Summary:** This script builds a Streamlit web app for user interaction, likely for health or chatbot purposes.

### ✅ **Advantages:**

- Reusable functions and modular code structure.
- Integration with modern libraries (Streamlit, Gemini LLM, PyPDF).
- Facilitates interactive AI-driven workflows.

## calories\_health.py (Food Calories Chatbot)

- This is a **Streamlit app** that uses **Google Gemini AI**.
- You upload a **food image**, and it analyzes the items in the image.
- The app asks Gemini to act like a **nutritionist** and:
  1. List each food item.
  2. Estimate **calories** for each item.
  3. Show the **total calories**.
- It works by:
  1. Taking user text input + uploaded food image.
  2. Sending both to Gemini AI.
  3. Showing the AI's response in a neat format.

👉 In short: **You upload a food picture → Gemini AI → Tells calories & food details.**

